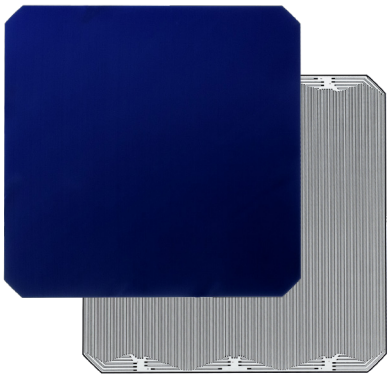
MAXEON® GEN 3

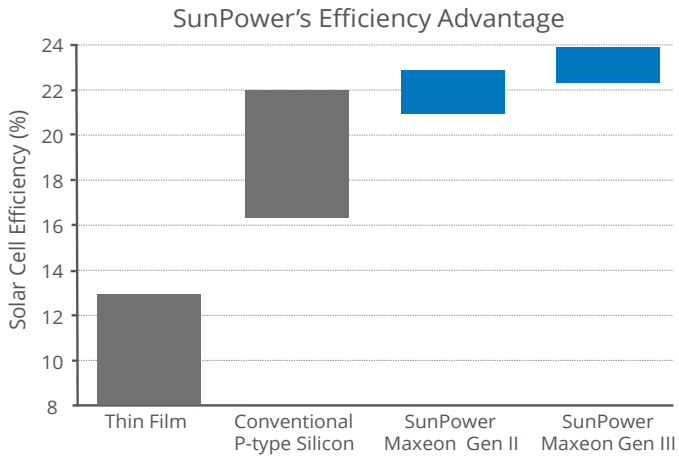
SUNPOWER®



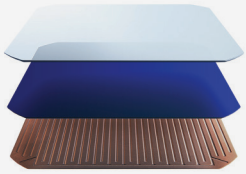
High Efficiency Solar Cell

Maxeon® Technology

Every day, entrepreneurs, designers, adventurers and explorers are changing the way our world is powered by placing their trust in Maxeon technology. We share your spirit of excellence and relentless innovation, which is reflected in most powerful and durable cell to solar enthusiasts. Together, even the boldest goals are within reach.



Fundamentally Different. And Better.



Maxeon® Technology

- Ultra-pure, n-type, monocrystalline silicon for maximum power
- Tin barrier prevents corrosion
- Uniquely durable back-contact design with no ribbons
- Clean and elegant aesthetics by designing out front contacts

Trusted Durability

- Solid metal foundation helps cell bend where others break under pressure.
- Conductive and malleable foundation keeps cell electrically intact even if eventually cracked.

Born to Break Records

Maxeon cells powered the first solar circumnavigation of the planet by air and by sea. They are the chosen technology of pioneers who demand the best in harsh environments.

Proven Technology Platform

SunPower has deployed more than one billion cells across more than 9 GW of installed solar - with a very low warranty return rate of 0.005%. SunPower's industry-leading R&D team has invested deeply in generations of incremental design enhancements over three decades.

Maxeon® Gen 3 High Efficiency Solar Cell

Electrical Characteristics of typical Maxeon Gen 3 Cell

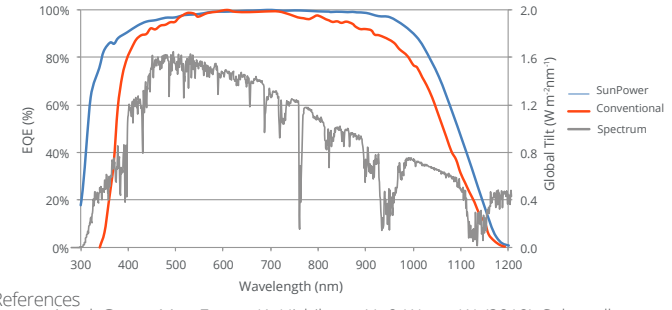
Cell Bins	Pmpp (Wp)		Eff (%)		Vmpp (V)		Impp (A)		Voc (V)		Isc (A)	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Ne3	3.77		24.34		0.625	0.627	6.050	6.054	0.731	0.734	6.382	6.412
Me3	3.49	3.76	22.50	24.33	0.617	0.624	6.038	6.052	0.725	0.732	6.374	6.397
Je3	3.14	3.48	20.27	22.49	0.603	0.608	5.568	5.803	0.723	0.733	6.197	6.285

Temp Coefficients in SunPower Panels: Voltage: -1.74mV/ºC, Current: 2.9mA/ºC, Power: -0.29%/ºC

Positive Electrical Grounding

If cell voltage is below frame ground the cell power output will be reduced. Therefore, modules and systems produced using these cells should be configured as "positive ground systems." If this creates a problem, please consul with SunPower

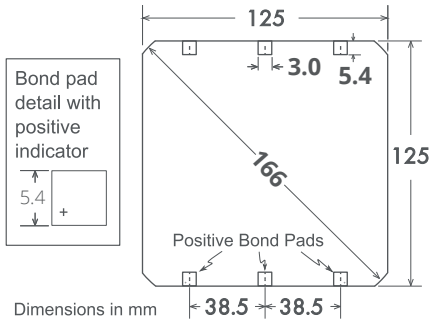
Spectral Response



References
Conventional: Green, M.a., Emery, K., Hishikawa, Y., & Warta, W. (2010). Solar cell efficiency tables (version 36). Progress in Photovoltaics: Research and Applications, 18(5), 46-352.doi:10.1002/pip/1021
SunPower: NREL data, commissioned by SPWR."Gen C CS AR binl". 2013.
Spectrum: Standard, ASTM. " G173-03." URL: http://www.astm.org

Cell Physical Characteristics

Wafer:	Monocrystalline silicon
Design:	All back contact
Front:	Uniform, black antireflection coating
Back:	Tin-coated, copper metal grid
Cell Area:	Approximately 155cm^2
Cell Weight:	Approximately 6.6 grams
Cell Thickness:	150µm +/- 30µm



Bond pad area dimensions are 5.4mm x 3.0mm
Metal finger pitch between positive and negative fingers is 471um
Positive/Negative pole bond pad sides have "+/-" indicators on leftmost and rightmost bond pads

Interconnect Tab and Process Recommendations

SunPower recommends customers use SunPower's patented tin-plated copper strain-relieved interconnect tabs, which can be purchased from SunPower. These interconnects are easily solderable and compatible with lead free processing.
Tabs weigh approximately 0.3 grams.

Our patented interconnect tabs are packaged in boxes of 1,200 each.

http://us.sunpower.com/about/sunpower-technology/patents/

Production Quality

ISO 9001:2015 certified

Soft handling procedures to reduce breakage and crack formation

100% cell performance testing and visual inspection

Packaging

Cells are packed in boxes of 1500 each; grouped in 10 shrink wrapped stacks of 150 with interleaving. 24 boxes are packed in a water-resistant "Master Carton" containing 36,000 cells suitable for air transport.

Purchase Terms

Customers shall not reverse engineer, disassemble or analyze the Solar Cells or any prototype, process, product, or other item that embodies Confidential Information of SunPower. Customers shall not cause or allow any inspection, analysis, or characterization of any properties (whether mechanical, structural, chemical, electrical, or otherwise) of the Solar Cells, whether by itself or by a third party.

Customer agrees that it will not transfer (whether by sale, loan, gift, or other conveyance) the Solar Cells from its possession.

SunPower solar cells are provided "AS IS" without warranty.

Full terms and conditions are in the Cell Purchase Agreement

SUNPOWER®

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